REPORT VISUAL QUESTION ANSWER TASK

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1. Introduction

This report has the objective to explain all the steps and methodologies applied throughout the process of completing the given task. Specifically, the given task was:

"Candidates should design and implement a vision question answering tool trained on the Fickr30k dataset (https://huggingface.co/datasets/nlphuji/flickr30k) . All-in-one multimodal LLM (like Qwen 3.5 and following) should not be used. Flamingo, CLIP, LLAVA or similar tools are welcome. The produced tool will be tested during the exam with an online demo where I will give to your system data it has not seen before."

So, the final goal of the task is to implement a model, which has to be trained on the Flickr30k dataset, that is able to perform the Visual Question Answering task.

2. Dataset and Model

As said in the previous paragraph, the dataset used to accomplish this task is **Flickr30k**, which contains about 31,000 images with 5 captions for each image. However, due to the lack of hardware power I was able to use only a small part of the entire dataset **(4000 images and only 1 caption for each image)**. This dataset is not natively created to be used to train a model on the VQA task, that is why I had to perform a further step, that I will explain later, to make it more suitable for this task. The type of images contained in this dataset is like the one on which multimodal models are trained on. In particular, the model that I decided to use is **LLAVA 1.5**. I chose this model because it should be more performative on this type of task than multimodal models like CLIP, which are better for tasks like Retrieval. Since LLAVA is a pretrained model, our goal was not to teach new things to it, but to adapt its knowledge to this specific dataset using fine-tuning techniques like LoRA.

3. Hardware Resources

The hardware resource used for the task is the GPU T4 x2. In particular, I used the GPU of Kaggle since it gives 30 free hours each week and each session can last until 12 hours straight. Another option could have been Google Colab, but it offers less hours of GPU and the maximum number of hours of a single session is about 8.

4. Analysis

4.1 Importing libraries and dataset

The first step I did for this homework was to import the necessary libraries and the Flickr30K dataset directly from Hugging Face.

4.2 Creating the VQA Dataset from Flickr30k

As said before, the Flickr30k is a dataset containing 31000 images and 5 captions for each image. This means that it is not possible to use it directly to train a model for the VQA task. To use a suitable dataset for the VQA task I used an LLM, specifically **DeepSeek V4 Flash**. Starting from 1 caption of an image, the LLM has to generate 5 question/answer pairs, which will be lately split between train and test set. The prompt I gave to the DeepSeek model is:

"Caption: "{caption}"

Generate exactly 5 diverse question-answer pairs based ONLY on this caption.

Cover different aspects: actions, locations, colors, objects, people, quantities.

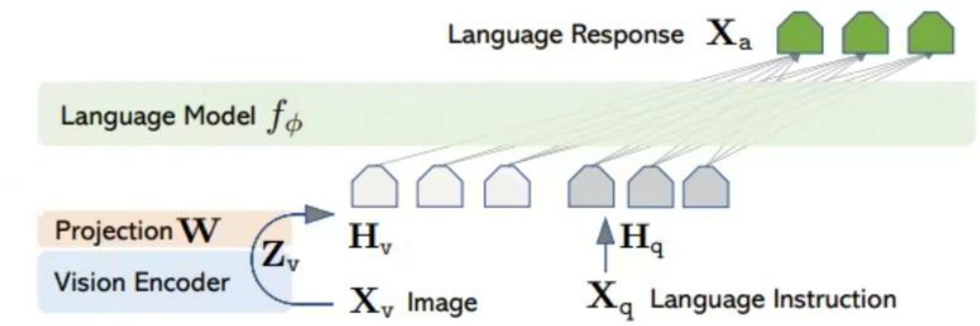
Do not hallucinate or invent details that are not present in the text.

Answers must be short (max 10 words) and directly answerable from the caption."

After running the script, I saved the train and test pairs into a Kaggle dataset, in this way this step was performed just once.

4.3 Loading LLAVA 1.5 Model and LoRA application

In this phase of the project, the core architecture is instantiated by loading the **LLaVA-1.5** (Large Language-and-Vision Assistant) model. From an architectural perspective, LLaVA operates as a highly cohesive multimodal pipeline: it leverages a pre-trained visual encoder (CLIP ViT-L/14) to extract dense visual features from images, a Multi-Layer Perceptron (MLP) projection module to translate and align these visual tokens into the text embedding space, and a robust Large Language Model (Vicuna-7B) to process the combined multimodal inputs and generate coherent responses.



LLaVA Architecture

Given the massive scale of this 7-billion-parameter architecture, performing a full fine-tuning on the available hardware (Kaggle dual T4 GPUs) would inevitably result in Out-Of-Memory (OOM) errors. To overcome this critical computational bottleneck, **LoRA (Low-Rank Adaptation)** was applied. Instead of updating the entire network, LoRA freezes the original pre-trained weights of the base model and injects small, trainable rank-decomposition matrices into the Transformer layers. This Parameter-Efficient Fine-Tuning (PEFT) strategy was deliberately chosen for two main reasons: it drastically reduces the VRAM requirements and training time, and it theoretically preserves the foundational multimodal reasoning capabilities of the base model while allowing for a targeted stylistic adaptation to the Flickr30k dataset.

4.4 First attempt of fine-tuning

For the initial training phase (Model V1), the system was configured using standard LoRA hyperparameters commonly found in text-only model fine-tuning literature. Specifically, the adapter was set with a LoRA Rank (r) of 16 and an alpha of 32, coupled with a relatively aggressive Learning Rate of 2e-4. Initially, the training metrics appeared highly promising: the loss function decreased steadily, and during inference, the model successfully demonstrated that it had learned the hyper-concise output style required by the Flickr30k dataset, generating accurate 2-3 word responses.

However, a deeper qualitative evaluation revealed a severe degradation in the model's underlying visual comprehension capabilities. By forcing the network to adapt its output format on a small subset of data (~4,000 images) using an aggressive learning rate, the model developed a severe overfitting to the textual syntax. To minimize the loss function rapidly, the network adopted a "minimum effort" strategy: it began to completely ignore the visual features extracted by the CLIP encoder, relying solely on text-based statistical probabilities. Empirically, this resulted in a loss of visual grounding. For example, when prompted to describe an image of a red pepper, Model V1 confidently responded with "a bow." This demonstrated that the large capacity of the LoRA adapter (r=16), driven by the high learning rate, overwhelmed the base model's pre-trained visual perception, rendering it functionally blind to the actual image content in its pursuit of stylistic conformity.

4.5 Second attempt of fine-tuning

To address the visual degradation observed in the first experiment, a second training phase (Model V2) was executed, shifting from an aggressive approach to a highly conservative parameter configuration. The goal was to enforce a strict mathematical bottleneck that would prevent the model from memorizing the dataset's textual syntax while simultaneously allowing it to learn the concise output rule.

To achieve this, the LoRA Rank (r) was halved from 16 to 8 (with alpha adjusted to 16), significantly reducing the capacity of the trainable adapter. This constrained the adapter space, forcing the network to extract the underlying general rule of brevity rather than overfitting the training samples. Furthermore, the Learning Rate was drastically reduced by an order of magnitude, from 2e-4 to 2e-5. This allowed the model weights to update gently and gradually, preserving the pre-trained visual pathways within the base architecture.

This targeted optimization yielded highly positive results. During inference, Model V2 successfully maintained the ultra-concise textual style learned from the Flickr30k dataset while successfully recovering its visual grounding capabilities. When presented with the same test images that triggered errors in V1, the optimized model accurately recognized objects, colors, and quantities, demonstrating a balanced integration of stylistic adaptation and sustained multimodal perception.

4.6 Evaluation

To objectively evaluate the performance of the optimized model (V2), a quantitative analysis was conducted using the **BERTScore** metric. Unlike traditional metrics based on exact n-gram matching (such as BLEU or ROUGE), BERTScore was specifically selected for its ability to assess *semantic similarity*. By leveraging pre-trained contextual embeddings, BERTScore calculates the semantic distance between the tokens generated by the model and those present in the dataset's ground truth. This methodological choice is particularly suited for Vision-Language tasks, where the primary objective is not the verbatim reproduction of a text string, but rather the conceptual and descriptive accuracy of the visual observations (e.g., recognizing "woman" and "girl" as semantically related terms).

The evaluation was performed on a representative sample of 5 examples randomly extracted from the validation set. In each analyzed instance, the model achieved an F1 score (which balances semantic Precision and Recall) **consistently close to or greater than 0.90**. This quantitative result empirically confirms the success of the parameter optimization: the model proved capable of generating descriptions that strictly adhere to the stylistic brevity constraints while remaining deeply accurate and semantically highly relevant to the actual visual content.

5. Conclusion

In conclusion, this project successfully demonstrated how to adapt a large Vision-Language Model (LLaVA-1.5) to a very specific and constrained output format using Parameter-Efficient Fine-Tuning (LoRA). The experiment highlighted a common and significant challenge in modern AI training: the risk of losing the model's original ability to correctly "see" and understand images when forcing it to learn a new textual style on a relatively small dataset. During the first training attempt, the model simply memorized the short text format to minimize the error, but it stopped paying attention to the actual visual details, leading to inaccurate and disconnected descriptions.

By systematically diagnosing the root cause of this issue—which was an overly aggressive learning rate combined with an adapter capacity (LoRA rank) that was too large—the training process was successfully adjusted. Lowering the learning rate and reducing the LoRA rank to 8 restricted the model's capacity to blindly memorize the text. This crucial adjustment forced the model to rely on its pre-existing visual intelligence while still learning the new rule of answering with just a few words. As a result, the optimized model successfully adopted the hyper-concise style of the Flickr30k dataset while keeping its foundational visual reasoning intact. Ultimately, this work emphasizes that fine-tuning is not just about feeding data to a network, but it requires careful parameter calibration: when balanced correctly, it becomes a highly effective tool to shape how a model communicates without breaking its original knowledge.

LINK TO THE KAGGLE NOTEBOOK:

https://www.kaggle.com/code/lorenzobove24/progetto-speech-processing